

# Classification

- ▶ In contrast to the regression problem where we would like to predict the value of a quantitative variable, we will now consider prediction of qualitative variables, i.e. which class an observation belongs to.



$$Y = \begin{cases} 1 & \text{with probability } p_1 \\ 2 & \text{with probability } p_2 \\ \vdots & \\ K & \text{with probability } p_K \end{cases}$$

- ▶ where  $p_1 + p_2 + \dots + p_K = 1$
- ▶ We want to predict  $Y_i$  based on  $\mathbf{X}_i$
- ▶ Models for this are usually based on estimating  $p_k$  as a function of  $\mathbf{X}_i$

$$p_k = P(Y_i = k | \mathbf{X}_i = \mathbf{x}_i) = f(\mathbf{x}_i)$$

## A natural quality measure of predictions

- ▶ Data:  $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n, \mathbf{x}_{n+1}, \dots, \mathbf{x}_{n+m}$   
 $y_1, y_2, \dots, y_n, y_{n+1}, \dots, y_{n+m}$



$\hat{y}_i$  = Prediction of  $y_i$

- ▶ Training error rate:

$$\text{trainER} = \frac{1}{n} \sum_{i=1}^n I(\hat{y}_i \neq y_i)$$

- ▶ Test error rate:

$$\text{testER} = \frac{1}{m} \sum_{i=n+1}^{n+m} I(\hat{y}_i \neq y_i)$$

- ▶  $I(x)$  is the indicator function (one if  $x$  true, zero otherwise)

## A first example

Credit card default data

```
require(ISLR)
```

```
## Loading required package: ISLR
```

```
summary(Default)
```

| ## | default  | student  | balance        | income        |
|----|----------|----------|----------------|---------------|
| ## | No :9667 | No :7056 | Min. : 0.0     | Min. : 772    |
| ## | Yes: 333 | Yes:2944 | 1st Qu.: 481.7 | 1st Qu.:21340 |
| ## |          |          | Median : 823.6 | Median :34553 |
| ## |          |          | Mean : 835.4   | Mean :33517   |
| ## |          |          | 3rd Qu.:1166.3 | 3rd Qu.:43808 |
| ## |          |          | Max. :2654.3   | Max. :73554   |

$Y_i = \text{Yes}$  if default and “No” otherwise

$X_i = \text{Yes}$  if student and “No” otherwise

```
##          default
```

```
## student    No    Yes
```

```
##      No  6850   206
```

```
##      Yes 2817   127
```

```
##          default
```

```
## student                No                Yes
```

```
##      No  0.97080499 0.02919501
```

```
##      Yes 0.95686141 0.04313859
```

►  $\therefore P(Y_i = \text{Yes} | X_i = \text{Yes}) > P(Y_i = \text{Yes} | X_i = \text{No})$

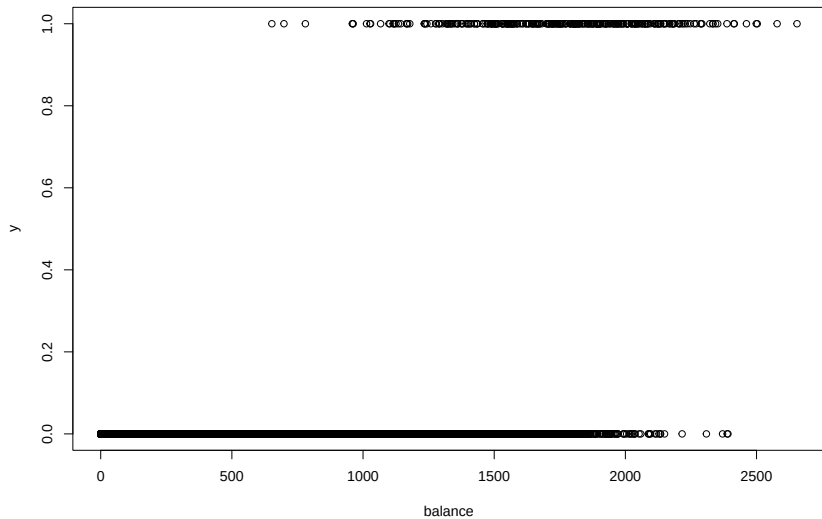
Recode the Y-variable so that

$Y_i = 1$  if default and zero otherwise

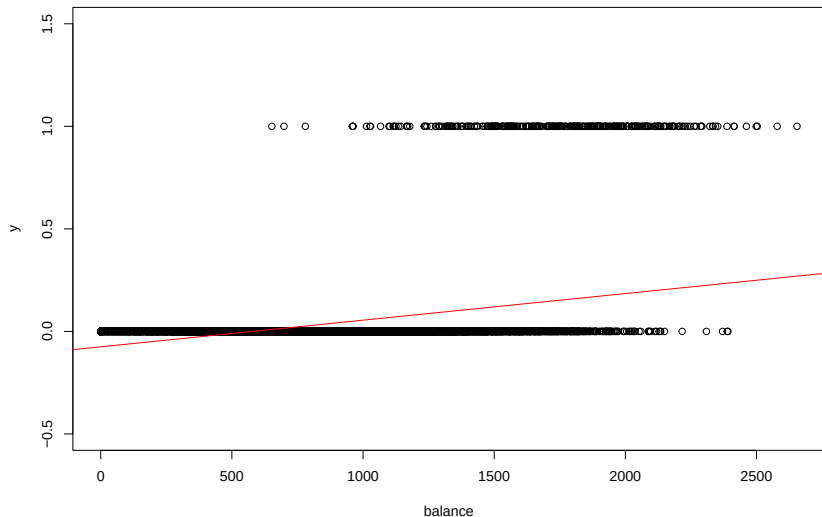
```
n=nrow(Default)
y=matrix(0,n,1)
y[Default$default=="Yes"]=1
table(student,y)
```

```
##           y
## student    0     1
##      No  6850  206
##      Yes 2817  127
```

# Account balance against default variable



# Our first classification method - a linear probability model



► Any obvious problems?

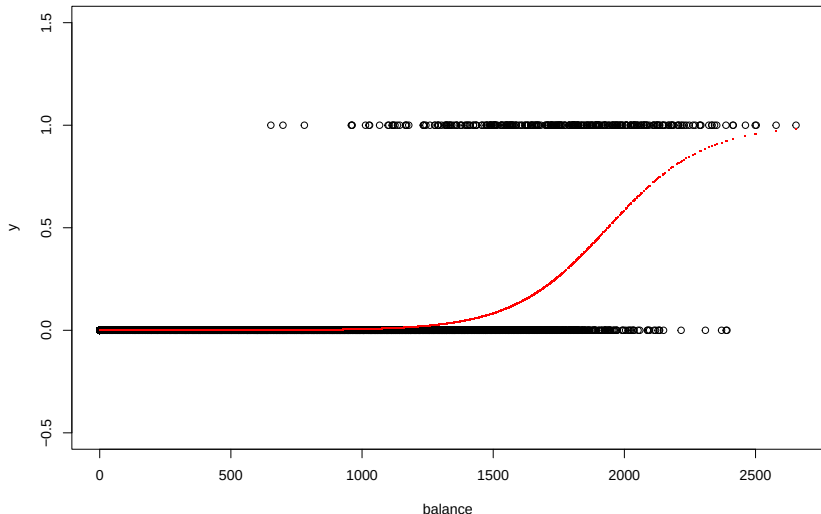
# Our first classification method - a linear probability model

```
linprob=lm(y~balance,data=Default)
summary(linprob)
```

```
##
## Call:
## lm(formula = y ~ balance, data = Default)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.23533 -0.06939 -0.02628  0.02004  0.99046
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -7.519e-02  3.354e-03  -22.42  <2e-16 ***
## balance      1.299e-04  3.475e-06   37.37  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1681 on 9998 degrees of freedom
## Multiple R-squared:  0.1226, Adjusted R-squared:  0.1225
## F-statistic: 1397 on 1 and 9998 DF,  p-value: < 2.2e-16
```



## Our first proper classification method - logistic regression



- Idea for prediction: Predict default when the estimated probability is larger than 0.5 (or some other threshold), i.e. when  $\text{balance} > 2000$ .

# Logistic regression

The prediction of  $Y$  is done in two steps

1. Estimate the probability of  $Y = 1$  given an observed value of  $X$ ,  $X = x$
2. Use the estimated probability to decide whether  $Y$  should be predicted as one or zero



$$P(Y = 1|X = x) = p(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}} = \frac{1}{1 + e^{-\beta_0 - \beta_1 x}}$$

- Note that  $\frac{p(x)}{1-p(x)} = e^{\beta_0 + \beta_1 x}$  is the *odds*, i.e. the probability of an event occurring in relation to the probability of the event not occurring.

This means that  $e^{\beta_1}$  represents the change in relative probability when  $x$  increases with one unit.

```
##
```

```
## Call:  glm(formula = y ~ student, family = "binomial", data = Default)
```

```
##
```

```
## Coefficients:
```

```
## (Intercept)    studentYes
```

```
##      -3.5041         0.4049
```

```
##
```

```
## Degrees of Freedom: 9999 Total (i.e. Null);  9998 Residual
```

```
## Null Deviance:      2921
```

```
## Residual Deviance: 2909  AIC: 2913
```

►  $\hat{p}(\text{not student}) = \frac{1}{1+e^{3.5041}}$

►  $e^{0.4049} = 1.50$

- Interpretation: The odds-ratio for default is 50% higher for students than for non-students

# Maximum likelihood (ML) estimation

## Example

$$Y = \begin{cases} 1 & \text{if a customer buys a product} \\ 0 & \text{if not} \end{cases}$$

The probability function for  $Y$  is

$$p(y) = P(Y = y) = \begin{cases} p & \text{if } y = 1 \\ 1 - p & \text{if } y = 0 \end{cases}$$

## ML estimation of $p$

- ▶ Random sample:  $Y_1, Y_2, \dots, Y_n$ , a sequence of independent random variables
- ▶ Observations (data):  $y_1, y_2, \dots, y_n$ , a sequence of numbers
- ▶ In this example  $m$  of the observations are one and  $n - m$  are zero
- ▶ The probability of observing the specific numbers  $y_1, y_2, \dots, y_n$  is

$$\begin{aligned}P(Y_1 = y_1, Y_2 = y_2, \dots, Y_n = y_n) &= [\text{indep}] \\&= P(Y_1 = y_1)P(Y_2 = y_2) \cdots P(Y_n = y_n) \\&= p^m(1 - p)^{n-m}\end{aligned}$$

- ▶ The likelihood function  $L(p) = p^m(1 - p)^{n-m}$
- ▶ The maximum likelihood principle: Find the value of  $p$  that maximizes the likelihood of observing the particular values of  $Y$  that we have actually observed.

Often easier to maximize the log-likelihood function, which in our example is

$$\ln L(p) = m \ln p + (n - m) \ln(1 - p)$$



$$\frac{\partial \ln L}{\partial p} = \frac{m}{p} - \frac{n - m}{1 - p} = 0$$

$$\Longleftrightarrow$$

$$m(1 - p) = p(n - m)$$

$$\Longleftrightarrow$$

$$\hat{p}_{ML} = \frac{m}{n}$$

- ▶ Check that second-order derivative  $< 0$ .

# ML estimation of parameters in logistic regression

Data:  $x_1, x_2, \dots, x_n$

$y_1, y_2, \dots, y_n$

- Probability of  $Y = y$  now a function of  $x$ .



$$p(y|x) = \begin{cases} \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}} & \text{if } y = 1 \\ \frac{1}{1 + e^{\beta_0 + \beta_1 x}} & \text{if } y = 0 \end{cases}$$

- Likelihood function:  $L(\beta_0, \beta_1) = p(y_1|x_1)p(y_2|x_2) \cdots p(y_n|x_n)$
- Can be shown that the log-likelihood function is

$$\ln L(\beta_0, \beta_1) = \sum_{i=1}^n \left[ y_i \ln \left( \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) + (1 - y_i) \ln \left( \frac{1}{1 + e^{\beta_0 + \beta_1 x_i}} \right) \right]$$

- Must be maximized numerically

# Evaluate accuracy

► Confusion matrix

|        |          | Prediction     |                |
|--------|----------|----------------|----------------|
|        |          | Negative       | Positive       |
| Actual | Negative | True negative  | False positive |
|        | Positive | False negative | True positive  |



## Default example

```
logprob=glm(default~student+balance+income,data=Default,  
             family="binomial")  
pred=predict(logprob,type="response")>0.5  
table(Default$default,pred)
```

```
##      pred  
##      FALSE TRUE  
## No    9627   40  
## Yes   228   105
```



|        |             | Prediction  |         |
|--------|-------------|-------------|---------|
|        |             | Not default | Default |
| Actual | Not default | 9627        | 40      |
|        | Default     | 228         | 105     |

# Evaluating the predictions on test data

```
set.seed(123)
n=nrow(Default)
ntrain=n/2
ind=sample(1:n,size=ntrain)
train=Default[ind,]
test=Default[-ind,]
logprob=glm(default~student+balance+income,data=train,
             family="binomial")
pred=predict(logprob,newdata=test,type="response")
conf_matrix=table(test$default,pred>0.5)
conf_matrix
```

```
##
##      FALSE TRUE
##   No   4809   20
##   Yes   116   55
```

```
accuracy=sum(diag(conf_matrix))/ntrain
accuracy
```

```
## [1] 0.9728
```

## Alternative prediction

- ▶ By classifying all 4839 testobservations as “Not default” we would get an overall prediction accuracy of  $4839/5000 \approx 0.9658$ .
- ▶ Overall prediction accuracy not a good measure when we have one class which is much larger than the other.